Submission Preview — Redacted Copy

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Nutrient-Specific Inequality in Global Food Availability: Energy, Protein, and Fat Across 176 Countries, 2020–2023

Submission ID: 2500760

Submission Type: Abstract – Student

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Abstract Details

Topic Area

1st choice: GI, Obesity, Metabolic, and Other Nutrition-Related Concepts

2nd choice: Malnutrition and Nutrition Assessment

Is your abstract fulfilling a grant reporting requirement for the ASPEN Rhoads Research Foundation annual grants program?

No

Background

Dietary energy availability is widely used to characterize food security at the population level, but energy-based indicators may obscure disparities in the availability of other nutritionally important macronutrients. Whether recent gains in mean food supply have been shared equally across populations, nutrient by nutrient, is directly relevant to population-level nutrition assessment. This study quantified cross-country inequality in energy, protein, and fat availability during 2020–2023 and evaluated whether increases in mean availability were accompanied by geographic convergence.

Methods

This ecological, country-level analysis used Food and Agriculture Organization of the United Nations (FAO) Food Balance Sheet data for 2020–2023 in a balanced panel of 176 countries with complete energy, protein, fat, and population data, representing 95.6% of the world population in 2023. Availability was expressed as kcal/person/day for energy and g/person/day for protein and fat; these values describe national food supplies available for human consumption, not individual dietary intake. The population-weighted Theil L index (mean log deviation) was the primary inequality measure, with population-weighted Gini coefficients as a sensitivity analysis. The analysis was descriptive; no inferential significance testing was performed.

Results

Between 2020 and 2023, population-weighted mean availability increased for all three nutrients: energy from 2,971.0 to 3,041.4 kcal/person/day (+2.4%), protein from 90.7 to 93.3 g/person/day (+2.9%), and fat from 86.8 to 90.7 g/person/day (+4.5%). In 2023, population-weighted Theil L was 0.0137 for energy, 0.0466 for protein, and

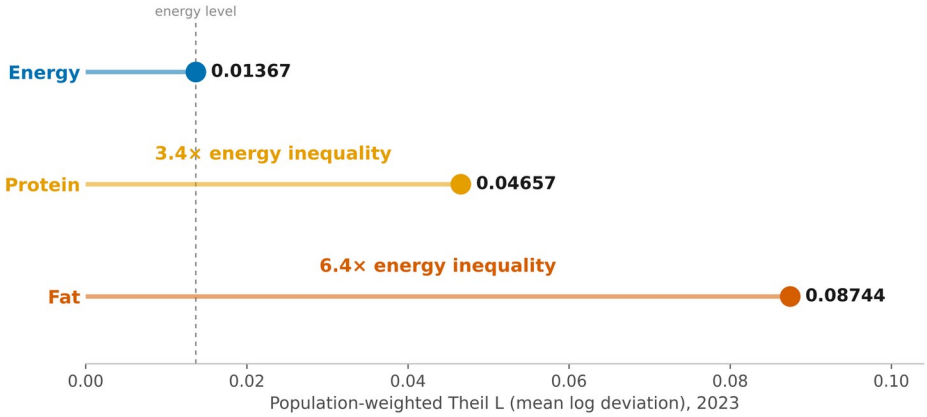
0.0874 for fat; cross-country inequality in protein and fat availability was therefore 3.4 and 6.4 times that in energy availability, respectively. Population-weighted Gini coefficients confirmed this ordering (energy 0.0918, protein 0.1632, fat 0.2279). Despite increases in mean availability, inequality did not uniformly decline: Theil L rose from 0.0124 to 0.0137 for energy (+10.0%) and from 0.0428 to 0.0466 for protein (+8.9%), whereas fat inequality decreased from 0.0884 to 0.0874 (-1.1%) while remaining the highest of the three nutrients in both years. Gini coefficients displayed the same temporal pattern, increasing for energy and protein and declining slightly for fat. Rising average nutrient supply was thus not accompanied by uniform cross-country convergence.

Conclusion

During 2020–2023, mean national availability of energy, protein, and fat all increased, yet protein and especially fat remained several times more unequally distributed across countries than energy, and inequality did not uniformly narrow. Calorie-based monitoring can therefore miss nutrient-specific disparities in national food availability. Population-level nutrition assessment should incorporate nutrient-specific measures and track the distribution as well as the average of national food supplies. These data describe national food availability, not individual dietary intake or nutritional status.

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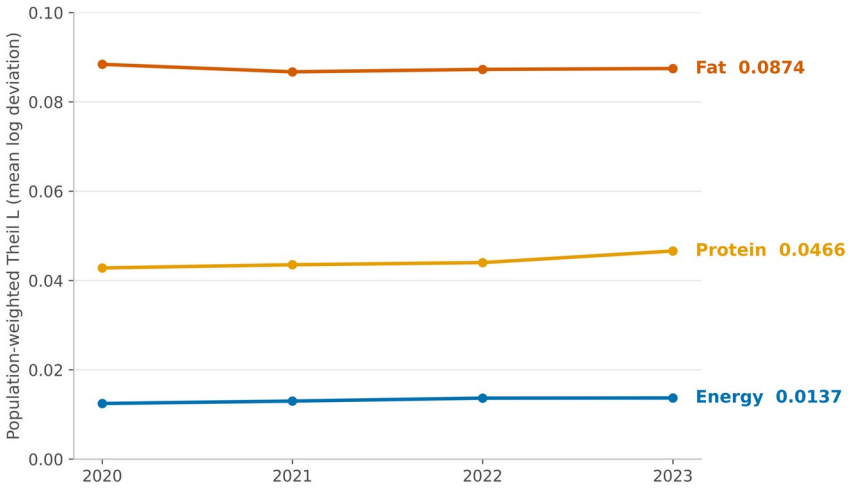
Images



Nutrient-Specific Inequality in National Food Availability, 2023

Population-weighted Theil L for national energy, protein, and fat availability across 176 countries. In 2023, cross-country inequality in protein and fat availability was 3.4 and 6.4 times that in energy availability, respectively.

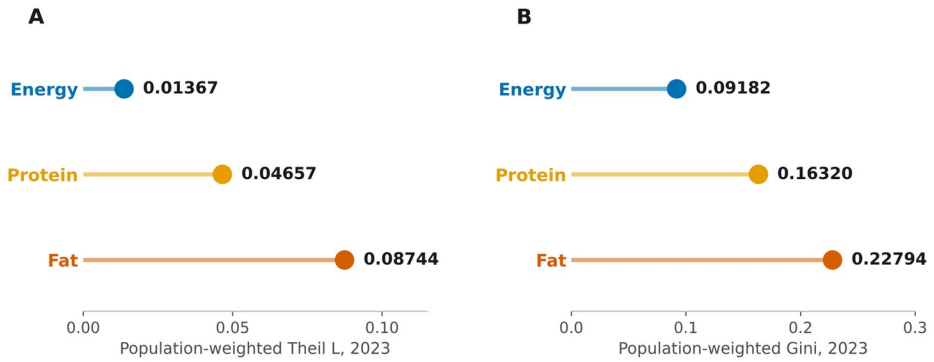
ASPEN27_Figure1_2023_Nutrient_Inequality.png



Cross-Country Inequality in Nutrient Availability, 2020–2023

Population-weighted Theil L for national energy, protein, and fat availability across a balanced panel of 176 countries. Fat availability remained the most unequal and energy availability the least unequal throughout 2020–2023.

ASPEN27_Figure2_Nutrient_Inequality_2020_2023.png



Sensitivity Analysis Using Theil L and Gini Coefficients

The ranking of national nutrient-availability inequality in 2023 was consistent across population-weighted Theil L and Gini coefficients: energy had the lowest inequality, followed by protein and fat.

ASPEN27_Figure3_Theil_Gini_Sensitivity.png

Presenter Photo and CV

[Redacted]

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This study used publicly available, de-identified, country-level aggregate data from FAO Food Balance Sheets and population datasets. No individual-level identifiable information, participant interaction, intervention, or collection of private human-subject data was involved; therefore, institutional review board approval was not required.

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Financial Support

PepsiCo, USD $3,500 research support

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